เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 — **Year:** 2012

Subject: 394172 Mathematics II — **Section:** 15-16

Date: 04 January 2012 — **Time:** 08.00-10.00

Name ______________________ ID ____________________ Class ______________

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks and dictionary are **NOT** allowed.
4. Calculators are allowed in the examination.
5. This is a closed book examination.
6. Electronic communication device are **Not** allow in the examination room.
7. The examination has 8 pages (including this page), 20 questions and 1 parts. The total score is 100 points.
8. Write all your answers on this examination sheet.

1. Construct a truth table for $[r \to (p \vee \sim q)] \leftrightarrow [(p \wedge \sim r) \to \sim q]$. (4 points)

2. Assume that p, q, r are **true** and s, t are **false**. Determine the truth value of

2.1 $[p \rightarrow \sim(q \rightarrow t)]$ (2 points)

2.2 $\sim[\, s \leftrightarrow (r \vee \sim t)] \rightarrow (p \rightarrow q)$ (3 points)

3. Is the proposition $[\, p \rightarrow (q \vee r)\,] \rightarrow [\,(p \wedge \sim q) \rightarrow r\,]$ tautology ? (4 points)

4. Assume that $A = \{-2, -1, 0, 1, 2\}$. Determine propositions truth value. (4 points)

4.1 $\forall x\ [\, |x| = x\,]$.

4.2 $\exists x\ [\, 2x^2 - 1 = x\,]$.

5. Determine the truth value of each proposition (8 points)

5.1 $\forall x \, \forall y \; [\, x + y^2 > 7 \,]$ where $A = \{3, 4, 5\}$

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5.2 $\exists x \, \exists y \; [\, x^2 + y^2 = 9 \,]$ where $A = \{0, 2, 3\}$

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5.3 $\exists x \, \forall y \; [\, x + y = y \,]$ where $A = \{0, 1, 2, 3\}$

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5.4 $\forall x \, \exists y \; [\, x + y \geq 0 \,]$ where $A = \{0, 1, 2, 3\}$

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6. Find the negation of "$\exists x \, [\, P(x) \wedge \sim Q(x) \,]$" ? (3 points)

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7.1 Ascending order of $3A5_{16}$, $9B0_{16}$, $BA0_{16}$, $32A_{16}$, $9AB_{16}$. (3 points)

7.2 Descending order of 1275_8, 10232_4, $21F_{16}$, 100010110_2. (3 points)

8. Express 3761 in base six. (3 points)

9. Express 536_8 is base 5. (4 points)

10. Express 312.02_4 in base 10. (4 points)

11. Evaluate $[\,3001_5 - 433_5] \div 3_5$. (Answer in base 5) (6 points)

12. Evaluate $[\,7552_6 + 5344_6] \times 24_6$. (Answer in base 6) (6 points)

13. Solve the solutions to the equations $x^4 - 5x^2 + 4 = 0$. (4 points)

14. Solve the solutions to the equations

$$\frac{(x+2)(x-3)}{5} - \frac{3x^2}{10} = \frac{3}{5}(x-1) - \frac{1}{10}(x-4)(x+3).$$ (5 points)

15. Solve the solutions to the equations $x^2 - 3x + \dfrac{40}{x(x-3)} = 14$. (5 points)

16. Solve $x^4 + 2x^3 - 13x^2 - 14x + 24 = 0.$ (5 points)

17. Solve (6 points)

$$\begin{aligned} 2x - 3y &= 4 \\ 2x^2 - 3xy - 2y^2 &= 4. \end{aligned}$$

18. Solve (6 points)

$$\begin{aligned} x^4 + x^2y^2 + y^4 &= 273 \\ x^2 - xy + y^2 &= 13. \end{aligned}$$

19. Solve (6 points)

$$\begin{aligned} 7x + 5y - 7z &= -8 \\ 4x + 2y - 3z &= 0 \\ 5x - 4y + 4z &= 35 \end{aligned}$$

20. Solve (6 points)

$$\begin{aligned} x^2 + y^2 + z^2 &= 84 \\ x + y + z &= 14 \\ xy &= z^2. \end{aligned}$$

ASST. PROF. DR. SUPHAWAT ASSAWASAMRIT